

**MATH 2101 LINEAR ALGEBRA I-PRACTICE PROBLEMS (UPDATED
15 NOV)**

1. EIGENVALUES AND EIGENVECTORS

- (1) Let A be an $n \times n$ matrix. Prove or disprove the following statements:
 - (a) λ is an eigenvalue for A if and only if λ is an eigenvalue for A^T .
 - (b) v is an eigenvector for A if and only if v is an eigenvector for A^T .
- (2) Let v be a non-zero vector in $\mathbb{R}^5$. Let A be an 5×5 matrix. Suppose $A^4v \neq 0$ and $A^5v = 0$.
 - (a) Show that $\{v, Av, A^2v, A^3v, A^4v\}$ forms a basis for $\mathbb{R}^5$.
 - (b) Let β be the ordered basis in (a). Find the matrix representation $[L_A]_\beta$. (Recall that L_A is the linear transformation given by $L_A(x) = Ax$.)
 - (c) Show that 0 is the only eigenvalue for A .
- (3) Determine if the following statements are true or false:
 - (a) Let A be an $n \times n$ matrix. If the geometric multiplicity of an eigenvalue of A is equal to n , then A is *always* diagonalizable.
 - (b) Let A be an $n \times n$ matrix. If the algebraic multiplicity of an eigenvalue of A is equal to n , then A is *always* diagonalizable.
 - (c) Let A and B be two $n \times n$ matrices. If A and B are diagonalizable, then AB is *always* diagonalizable.
- (4) Determine if the following 6×6 matrices are diagonalizable:
 - (a) $\begin{pmatrix} I_3 & I_3 \\ 0 & I_3 \end{pmatrix}$;
 - (b) $\begin{pmatrix} I_3 & I_3 \\ 0 & 2I_3 \end{pmatrix}$.
- (5) Diagonalize the following matrices:
$$\begin{pmatrix} 3 & 0 & 0 \\ 1 & 3 & 1 \\ -1 & 0 & 2 \end{pmatrix}.$$
- (6) Let A be an $n \times n$ matrices. Suppose A has only two distinct eigenvalues λ_1 and λ_2 . Prove that A is diagonalizable if and only if for any $v \in \mathbb{R}^n$, $v = v_1 + v_2$ for some $v_1 \in E_{\lambda_1}$ and $v_2 \in E_{\lambda_2}$.